What a Benchmark Score Cannot Say: Variance Components and Detectability Limits on Three Agent Benchmarks

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Abstract

When two agents differ by a few points on a public benchmark, is that a capability difference or measurement noise? We answer with a pre-registered measurement study of the three public agent benchmarks whose repeated runs survive artefact-level ingest verification (94,555 run outcomes; three of six candidates failed verification and were excluded): we decompose score variance into task-sampling and run-level components with two co-primary estimators — method of moments and a Bayesian hierarchical model — under a frozen rule for reporting their disagreement, and we convert the components into per-benchmark minimum-detectable-effect (MDE) design tables and an audit of published gaps on the two benchmarks that publish auditable ones. Run-level variance is 0.38–0.71 of task-sampling variance on the two benchmarks where the pre-registered verdict is readable, and is indeterminate, not small, on the third (a two-agent benchmark); the two estimators disagree on the level by 0.09–0.11 throughout, a disagreement we report rather than average away, and pooled ratios summarise wide per-agent spreads. Detectability, not repetition, is the binding constraint: on the METR task suite a five-point gap cannot be resolved by any number of runs at the headline design and needs 257 tasks at eight runs, while both of τ-bench's official leaderboard gaps fall below the minimum detectable effect of the design that produced them. On the benchmarks the public record lets anyone measure, a score as currently reported does not carry the information needed to say whether the differences it advertises are real.

Keywords: agent evaluation; benchmark metrology; variance components; minimum detectable effect; reproducibility

1 Introduction

Leaderboards rank agents by point scores. A two- or three-point gap between adjacent entries is routinely read — in papers, in model cards, and in procurement decisions — as a capability difference. That reading assumes the score is a precise measurement of the agent. It is not obviously so: an agent benchmark score is a mean over a finite set of tasks, each attempted a small number of times by a stochastic system, and every one of those finitudes contributes variance that the published number does not report. One live industry leaderboard's methodology page states the position plainly: its intervals "reflect within-run case-level sampling, not pass-to-pass stability", and "run-to-run variance is not yet measured for this board" [20]. The field's own systematic survey reaches the same conclusion at scale: across 259 studies, non-determinism is named an uncontrolled and unmeasured threat to benchmark conclusions [2].

The components of that unmeasured variance have each been touched somewhere. Miller [8] gives the reference treatment of task-sampling uncertainty for static question-answering evaluations, with paired comparisons and power analysis, but no run, grader, or environment component

and nothing agentic. Bjarnason et al. [4] measure run-to-run spread on one benchmark (single-run pass@1 varies by 2.2–6.0 points across runs) and recommend power analysis, but decompose nothing and audit nothing. Zhou et al. [21] decompose variance on one static code benchmark into two components and derive a repetition ceiling; Messing [22] decomposes the uncertainty of LLM annotation pipelines, including interactions, but not of repeated agent runs. The machinery we use is standard experimental-design machinery — ANOVA-style variance components in the line of Bouthillier et al. [10] and generalizability theory, and power analysis in the line of Card et al. [12] and Miller [8] — and we claim no novelty for it. What none of the works above provides — and what a reader of a leaderboard actually needs — is the three-part conjunction: a decomposition of an agent benchmark's score variance on genuine repeated runs, the conversion of those components into the smallest gap the benchmark's own design can detect, and an audit of the published record against that detectability limit. This paper provides that conjunction, under a pre-registration frozen before any census data was touched, on every public benchmark whose repeated runs survive verification at the artefact itself.

The contributions are:

1. **A variance census with two co-primary estimators and a frozen disagreement rule.** Across the three benchmarks that pass ingest verification, the ratio of run-level to task-sampling variance is $\rho = 0.3767$ [0.3154, 0.4426] (METR task suite) and 0.7134 [0.6355, 0.7844] (SWE-bench Verified repeated-run corpus) under the moment estimator, with τ-bench indeterminate at 0.9834 [0.7798, 1.2239]. The Bayesian arm reads 0.09–0.11 higher on all three and is formally discordant with the moment arm throughout; both arms are reported in full, and the verdicts are read from the arm the frozen threshold was defined on. Estimator disagreement at this size is itself a measurement result: it bounds how precisely the field's current tools can state the noise level of a benchmark.

2. **Detectability tables, including the regime where runs cannot help.** Per benchmark, the minimum detectable effect at 80% power as a function of runs and tasks, with required-runs and required-tasks inverses. The headline case is an impossibility, not a shortfall: where the task term binds, no number of runs resolves the gap, and the design table must say so.

3. **An audit of published gaps against their own designs.** Both gaps on τ-bench's official leaderboard are smaller than the minimum detectable effect at the design actually used; of 24 published contrasts on the SWE-bench Verified corpus, 18 clear their design's MDE, 6 fall below it, and 3 change classification with the estimator. One benchmark contributes no auditable gap at all, because it publishes none with a stated design — a finding about reporting practice, reported as such.

A note on what this paper is not. It is a measurement of published benchmark designs and records, not a proposal for a new benchmark, not a validity or shortcut analysis (a different object: bias, not variance [17, 18]), and not a re-analysis of any single benchmark's capability claims. Its negative results — an indeterminate benchmark, an empty audit leg, three excluded corpora — are reported at the same prominence as its positive ones, because under a pre-registered design they are part of the answer.

2 The measurement problem

A published score for agent i is a mean over a finite design, $\hat{S}_i = \frac{1}{T}\sum_j \frac{1}{R_j}\sum_r Y_{ijr}$, where j indexes tasks drawn from the benchmark's task set (size T), r indexes independent runs (count R_j), and

Component	Definition (per agent i)	Generated by
Task sampling $\sigma^2_{\text{task}}(i)$	$\text{Var}_j(p_{ij})$	Which tasks are in the benchmark
Run-level $\sigma^2_{\text{run}}(i)$	$\text{E}_j[\text{Var}_r(Y \mid i, j)]$	Agent stochasticity; environment non-determinism where inseparable
Grader $\sigma^2_{\text{grader}}(i)$	$\text{E}[\text{Var}(Y \mid \text{frozen trajectory, re-grading})]$	Stochastic or model-based grading
Environment $\sigma^2_{\text{env}}(i)$	$\text{E}[\text{Var}(Y \mid \text{frozen policy and grading})]$	Live-environment drift

Table 1: The components of a benchmark score's variance. The environment component is counted within the run-level component wherever a corpus cannot identify it separately (refinement R1, Section 3); the grader component is estimated only where a corpus re-grades frozen trajectories, and is otherwise an estimability limit, never an imputed zero.

Y_{ijr} is the run outcome — binary as the primary endpoint, continuous partial credit as secondary. Under the random-effects model $Y_{ijr} = p_{ij} + \varepsilon_{ijr}$, with p_{ij} the agent's per-task success propensity, the law of total variance splits a single run's outcome variance, for fixed agent i, into named components (Table 1). At design level, the variance of the published number is approximately $\text{Var}(\hat{S}_i) \approx \sigma^2_{\text{task}}/T + \sigma^2_{\text{run}}/(T \cdot R)$ (plus a grader analogue where identified): tasks and runs buy precision at different rates, and only the first term responds to a larger benchmark while only the second responds to repetition. A leaderboard that reports neither component cannot say which lever its own noise responds to — or whether either lever suffices for the gaps it ranks.

The falsifier statistic is the ratio $\rho = (\sigma^2_{\text{run}} + \sigma^2_{\text{grader}})/\sigma^2_{\text{task}}$, pooled across agents with equal weights, per benchmark. In every corpus this census could measure, σ^2_{grader} is zero by construction (programmatic grading, no re-grades), so the estimated numerator is the run-level component alone; ρ is written in its general form because the estimand, frozen before the census, is defined for judged benchmarks too — and on judged benchmarks it remains unmeasured, here and, to our knowledge, everywhere (Section 6). The thesis of this paper — that single-run scores are not as reliable as the field's usage assumes — dies, by the pre-registered criterion, if $\rho < 0.10$ on a set of at least three benchmarks with genuine repeated runs. The threshold is deliberately ungenerous to the thesis: run noise one-tenth the size of task noise would already vindicate current practice.

3 Design: a pre-registered census

3.1 Freezing, and what was frozen

The full design — estimands, selection rule, falsifier, estimators, diagnostics, and the disagreement rule below — was frozen on 2026-10-31 (addendum timestamp 2026-10-31T16:00:07Z) in a SHA-256 hash chain, before any census data was downloaded, ingested, or computed on, and before the model-based arm's code existed. The bounded pilot that preceded it (one agent on one benchmark, $\rho = 0.245$, 95% CI [0.134, 0.397]) was itself run under a pre-registration frozen 2026-10-30; its reading is indicative only, since the falsifier's verdict requires the full set. Two clarifications were approved with the investigation packet and carried into the freeze verbatim: **R1** — where a corpus cannot separately identify the environment component, it is counted within the run-level component and the inseparability reported as an estimability limit; the same treatment applies to the grader component where no re-grading exists (it is then either zero by construction, for deterministic programmatic grading, stated as such, or not identified). Amendments to any frozen quantity would have required a new frozen amendment before the affected outcomes were examined; none was made.

Benchmark	Outcomes	Agents	Tasks	Cells $n \geq 2$
B1: METR task suite (HCAST + RE-Bench)	39,363	52	177	0.979
B3: τ-bench (retail + airline)	1,980	2	165	1.000
B4: Bjarnason et al. corpus (SWE-bench Verified)	53,212	12	500	0.991

Table 2: The census set. B4's agents are agent configurations (two scaffolds × three models × two temperatures, not all cells present). B1 has 51 estimable agents; one (gpt2) has no repeated cells and is excluded by the $n_{ij} \geq 2$ rule. In B1, 108 identical duplicate run records were counted once and 308 conflicting re-listings excluded, both logged in the ingest ledger.

3.2 The census set: selection and ingest verification

A benchmark enters the set if and only if (a) genuine repeated runs ($n_{ij} \geq 2$ for a substantial share of agent×task cells) are publicly available at run level, (b) outcomes are binary or binarisable without judgement calls, and (c) task identities are stable across runs. Six candidates were carried into the census. Each was then subjected to *ingest verification*: the outcome-bearing files must be enumerated and parsed at the artefact itself, their counts reconciled against the corpus's stated design, and the licence permitting research reuse confirmed. Repository existence is not data presence — a rule this programme learned before the census, when a headline 2026 result in this exact field advertised a public outcome tensor that its repository does not contain at any ref (the kill is recorded in the replication record; that work's numbers are cited in this paper only as its authors' claims, never as verified).

Three candidates pass (Table 2); three fail, and the failures are findings:

- **B2 (HAL generalist traces): FAIL.** The dataset returns HTTP 401 at census time across its page, file tree, and file-resolve endpoints; the outcome files (encrypted archives) can be neither enumerated nor parsed.
- **B6 (HAL CORE-Bench): FAIL.** The same access failure; its frozen "re-verified at census" repeat-structure check is therefore impossible.
- **B5 (de la Rubia & Pafka coding-agent runs): FAIL on the selection legs.** The corpus is present and MIT-licensed (468 run records), but its outcome is a continuous holdout AUC with no pre-stated binary bar, and it contains exactly one task, so the task component has no population to estimate over.

The passing set is exactly the pre-registered floor of three. That thinness is itself a result about the field's data practices: of six candidate benchmarks chosen *because* they advertise repeated runs, half cannot support the measurement, two of them because advertised-open data is not, at the artefact, open.

3.3 Anchors

Before any variance estimate from a benchmark is trusted, a published-style score is recomputed from the raw run-level records two independent ways, with a record-count cross-check. B1's estimates reproduce the sealed pilot artefact bit-for-bit on overlapping slices (maximum absolute difference 0.0); B4's per-configuration means reproduce the source artefact to 1.4×10^{-14}; B3's two published contrasts are reproduced by the census corpus itself (its trajectories are the leaderboard runs). One discrepancy is on record: the pilot document's printed slice table is not reproducible

from its sealed artefact; the census anchors on the artefact, and the printed table stands as an uncorrected reporting error in the frozen document. The frozen thresholds are design constants and are unaffected.

3.4 Two co-primary arms, and the disagreement rule

Arm A (method of moments) estimates the components without distributional assumptions: within-cell variance $s_{ij}^2 = \frac{n_{ij}}{n_{ij}-1}\hat{p}_{ij}(1-\hat{p}_{ij})$, averaged with n_{ij} weights for σ^2_{run}; σ^2_{task} by the standard ANOVA moment correction (observed spread of cell means minus its expected sampling inflation, truncated at zero with the truncation reported). Uncertainty is by cluster bootstrap over tasks ($B = 2{,}000$), with BCa intervals reported throughout. One methodological deviation is on record here rather than repaired silently: the frozen registration specifies percentile bootstrap intervals; the census reports BCa. Both interval types are deposited in the artefacts; the fresh-tester audit of this census verified that the percentile intervals yield identical verdicts and identical disagreement-rule outcomes, so no finding in this paper depends on the substitution. **Arm B (Bayesian partial pooling)** fits, per agent, a logistic random-effects model $\text{logit}(p_{ij}) = \mu_i + v_{ij}$, $v_{ij} \sim \text{Normal}(0, \tau_i)$, with priors $\mu_i \sim \text{Normal}(0, 2.5^2)$ and $\tau_i \sim \text{HalfNormal}(0, 1)$, and derives the same probability-scale estimands per posterior draw, so both arms estimate the same quantities. A continuous-outcome arm (LMM on the original scale) is secondary in both. Sampler: NUTS, 4 chains, 2,000 warm-up + 2,000 draws, target acceptance 0.95. Diagnostics ($\hat{R} \leq 1.01$, bulk/tail ESS ≥ 400, zero divergences, posterior predictive checks) carry frozen consequences: exactly one remediation at stricter settings, then FAILED, recorded and never dropped. A pre-specified prior-sensitivity refit (widened priors) is always reported alongside.

Neither arm may be picked after results are seen, so the freeze specifies the **disagreement rule** in advance. The arms materially disagree on a benchmark if (a) the pooled ρ points differ by more than 0.05, (b) the two 95% intervals lie on opposite sides of the 0.10 falsifier threshold, or (c) a pooled E1/E2 point estimate differs by a factor outside $[2/3, 1.5]$. On material disagreement: both arms are reported side-by-side in full, the benchmark is flagged DISCORDANT, and the formal verdict is read from Arm A alone — because the falsifier, its threshold, and R1 were frozen in moment-estimator terms, and re-basing a frozen threshold onto a model-based scale would move the bar. Criterion (c), we record here, turned out not to be evaluable for this census: the frozen Arm B pooled artefact carries ρ only, with no pooled components, so no factor is computed rather than estimated post hoc. That is a limitation of the frozen artefact set, stated rather than repaired.

3.5 Detectability and the audit

For two agents compared on the same task set (the paired design leaderboards actually use), $\text{SE}_{\text{diff}}(R) = \sqrt{\sigma^2_{\text{task,diff}}/T + (\sigma^2_{\text{run}}(A) + \sigma^2_{\text{run}}(B))/(T \cdot R)}$ and the minimum detectable effect at two-sided $\alpha = 0.05$, power 0.80 is $\text{MDE}(R) = 2.802 \cdot \text{SE}_{\text{diff}}(R)$. The inverse, $R_{\text{required}}(\delta)$, has an honest edge case: if its denominator is non-positive, *no number of runs suffices* and the table reports T_{required} at the benchmark's actual R instead. That edge case is a result, not an error. Arm B's MDE construction is unpaired (its frozen model carries no per-pair estimand); both arms' tables are reported, and rows whose audit classification flips between arms are flagged arm-sensitive, never resolved by choosing an arm. The audit (E6) classifies each published pairwise gap that can be sourced to the benchmark's own publication or official leaderboard — with the gap, and the T and R of the design actually used, recorded — against the MDE at that design. No gap is reconstructed, estimated, or invented where no published value can be sourced. The audit is deep rather than

Benchmark	Arm A ρ (BCa 95%)	Arm A	Arm B ρ (95% CrI)	Arm B
B1 (METR)	0.3767 [0.3154, 0.4426]	SUPPORTED	0.4709 [0.4537, 0.4886]	SUPPORTED†
B3 (τ-bench)	0.9834 [0.7798, 1.2239]	INDET.	1.0823 [0.9200, 1.2741]	INDET.
B4 (Bjarnason)	0.7134 [0.6355, 0.7844]	SUPPORTED	0.8237 [0.7960, 0.8526]	INDET.‡

Table 3: Falsifier verdicts, both arms. †B1 Arm B carries the prior-sensitivity flag (Section 4.2). ‡B4 Arm B's indeterminacy is a fit-failure verdict under the frozen rule (Section 4.3), not an evidence verdict; its pooled point covers the 9 agents whose primary fits passed. Formal verdicts are read from Arm A alone under the frozen disagreement rule; all three benchmarks are DISCORDANT on criterion (a).

broad by design: it asks of each gap the strongest question (does it clear its own benchmark's detectability limit?) over few benchmarks, rather than a weaker question over many.

4 Results

4.1 The falsifier verdicts

Table 3 gives both arms, side-by-side as the frozen rule requires. The formal verdicts, read from Arm A: **B1 SUPPORTED** ($\rho = 0.3767$ [0.3154, 0.4426]); **B3 INDETERMINATE** ($\rho = 0.9834$ [0.7798, 1.2239]); **B4 SUPPORTED** ($\rho = 0.7134$ [0.6355, 0.7844]). No benchmark is refuted: wherever the verdict is readable, run-level variance is between roughly two-fifths and seven-tenths of task-sampling variance — four to seven times the level at which the thesis would have died. B1's continuous secondary endpoint corroborates its binary primary ($\rho = 0.4350$ [0.3657, 0.5102]).

B3's indeterminacy deserves plain statement, because it is the kind of result a less disciplined write-up would round into support. With two agents, the pooled estimate rests on a cross-fitted pair ratio; its point estimates sit at or above $\rho = 1$ — run variance comparable to or larger than task-sampling variance — but the frozen SUPPORTED clause requires the interval to lie entirely inside the definitional range (0, 1), which intervals crossing 1 fail. The frozen rules therefore record indeterminate. That is a precision statement, not a weak effect, and it is also a statement about τ-bench's published design: a two-agent repeated-run corpus cannot certify the quantity its leaderboard most needs. Per-agent readings underline the spread the pooled numbers summarise: within B1, per-agent ρ runs from 0.15 to 15.6 (median 0.35; the maximum is a near-degenerate low-task-variance agent, reported in the artefacts, not trimmed), and B3's two agents read 0.93 and 1.05 individually.

The grader component is an estimability limit throughout: no corpus in the set re-grades frozen trajectories, and grading in all three is programmatic, so E3 is zero by construction in all three — stated, not estimated. Nothing in this census measures grader variance on judged benchmarks; Section 6 returns to this.

4.2 The disagreement, reported

The disagreement rule fired on criterion (a) for all three benchmarks: $|\Delta\rho| = 0.0942$ (B1), 0.0989 (B3), 0.1103 (B4). Criterion (b) fired nowhere — every interval in both arms lies entirely above the 0.10 threshold, so the arms never disagree about which side of the falsifier the evidence sits on. They disagree about the level, by about a tenth of the task component, consistently in one direction (the model arm reads higher). We do not resolve this. The registration names candidate causes —

Benchmark (headline T)	R=1	R=2	R=4	R=8	R_{actual}	δ=0.05 needs	δ=0.02 needs
B1: METR (T=163)	0.1045	0.0832	0.0702	0.0627	0.0605 (R≈10.9)	257 tasks @ R=8	1,603 tasks @ R=8
B3: τ-bench (T=165)	0.1243	0.0981	0.0820	0.0725	0.0758 (R=6)	348 tasks @ R=8	2,171 tasks @ R=8
B4: Bjarnason (T=500)	0.0461	0.0328	0.0236	0.0171	0.0156 (R≈9.9)	R=1	R=6

Table 4: Minimum detectable effects (score points, normal-critical values; t-critical values are within 0.001 throughout and change no verdict) for the headline pair of each benchmark, paired design, Arm A. "Needs" columns give the binding resource: where runs cannot reach δ at the headline T, the required task count at $R = 8$; where they can, the required runs at the headline T. A 10-point gap needs only $R = 2$ on B1 and B3.

sparse cells, prior influence at the sensitivity refit, model misfit — and forbids post-hoc resolution; what can be said from the record is where each flag sits. B1's Arm B carries the prior-sensitivity flag: widening the priors moves its pooled ρ_B by 0.0748, above the 0.05 flag threshold. The caveat that travels with that flag is that the widened-prior pool covers only 28 of 51 agents (23 sensitivity fits failed the frozen diagnostics), so the shift mixes a prior effect with an agent-set effect, and the frozen construction does not decompose the two. B3's corresponding shift is 0.0499 — below the flag by 0.0001, recorded as borderline, not flagged. B4's is 0.0015, no flag. For a reader who wants one number: the honest one-number summary of this census is not a number but a range with a rule — the moment arm's reading, the model arm's reading 0.09–0.11 higher, and the pre-registered reason the formal verdict follows the former.

4.3 Fit accounting

Arm B's frozen list is 297 fits. Final tally: **246 PASS, 12 REMEDIATED_PASS, 39 FAILED, 0 FLAGGED**. Every failure followed the frozen ladder exactly and is recorded in its fit file with first-attempt and remediation diagnostics. Two clusters matter. In B1, 23 of 51 widened-prior sensitivity fits failed (Section 4.2), and 2 of 51 primary fits failed (davinci_002, gpt_3_5_turbo_instruct — neither a headline agent; 3.9% < 25%, so the Arm B verdict stands; misfit share 0.0196). In B4, the primary fits for *both* headline-pair configurations (Devstral-2 at temperatures 0 and 0.2) and for Qwen3-32B at temperature 0 failed on bulk effective sample size (359.6 falling to 277.7 after remediation, against the 400 bar; $\hat{R} \approx 1.005$–1.007; zero divergences throughout). Under the frozen headline-failure rule, B4's Arm B verdict is therefore indeterminate regardless of its pooled point. We regard the refusal to rescue these fits — no third attempt, no relaxed diagnostic, no model change — as part of the method, not an execution shortcoming: a metrology paper that quietly re-tunes its own instrument until the readings agree would be making the field's mistake at one remove.

4.4 Detectability: what the designs can resolve

Table 4 gives the moment arm's paired-design MDEs for each benchmark's headline pair; the model arm's unpaired tables (uniformly larger, since they cannot exploit shared tasks) are in the deposited artefacts and summarised in the text where they change a conclusion.

Read B1's row as the paper's central practical claim. Going from one run to the ≈11 runs actually used buys a factor of 1.7 in detectability (0.1045 to 0.0605) — and then the curve is spent: at $R = 8$ the MDE is 0.0627, and no further repetition moves it materially, because the remaining variance is the task term, which repetition does not touch. A five-point gap — larger than most adjacent leaderboard gaps — is simply not resolvable on this design; resolving it is a

task-set decision (257 tasks at $R = 8$), not a compute decision. The statement is a consequence of the measured components under the stated paired design model, not an empirical impossibility claim about METR scores in general; and because the two arms' MDE constructions differ (Arm A's is paired, Arm B's unpaired), their tables are not two estimates of one quantity — Arm B's larger values are what the same components imply when shared tasks cannot be exploited, and the audit reports both rather than blending them. B3 is worse in the same way (348 tasks needed, on a 165-task benchmark). B4 inverts the pattern for an instructive reason: its headline pair is one model at two temperatures, whose task-difference variance is 0.0020 — two orders below B1's 0.0609 — so runs bind and small gaps are reachable ($R = 6$ for two points). Detectability is a property of the pair and the design, not of the benchmark's size or budget; it has to be computed, per comparison, from measured components. At present it is computed by no one.

4.5 The audit

B3 (2 rows). τ-bench's official leaderboard reports pass[1] for gpt-4o and claude-3.5-sonnet on retail (0.604 vs 0.692, gap 0.088) and airline (0.420 vs 0.460, gap 0.040), from designs of $T = 115$, $R = 4/8$ and $T = 50$, $R = 4/8$ respectively (the leaderboard values equal the census corpus's own pass[1]: these are the leaderboard runs). The MDEs at those designs are 0.0927 and 0.1412. **Both published gaps are below the minimum detectable effect of the design that produced them** — under the moment arm, and identically under the model arm (MDEs 0.1383 and 0.2097), so the classification is not arm-sensitive. The retail gap misses its MDE by 0.005: close enough that a slightly larger trial count would have resolved it, which is precisely the kind of statement a design table exists to make before, not after, publication.

B4 (24 rows). Of the 24 contrasts in the source paper's Table 1, at the design actually used ($T = 500$, $R = 10$): 18 are above their MDE under both arms; 6 are below under both — four near-zero temperature contrasts (gaps 0.000–0.016 against MDEs 0.0148–0.0202), a same-temperature model pair on one scaffold (0.005 vs 0.0359), and a cross-scaffold gap (0.012 vs 0.0195); and 3 are arm-sensitive, above under the moment arm and below under the model arm (gaps 0.030–0.040). The corpus's headline claims — model differences of 0.43–0.47 — clear their MDEs by an order of magnitude. A benchmark whose own authors publish repeated runs is, on this evidence, a benchmark whose headline claims mostly survive their measurement uncertainty; the failures concentrate exactly where a reader would hope an audit looks: in the small contrasts a table invites but its design cannot support.

B1 (0 rows). No pairwise gap could be sourced to METR's own publications with both a numeric value and the design's (T, R): METR publishes time horizons and bucketed success rates, not overall pairwise gaps at a stated design. No gap was reconstructed (the frozen rule forbids it). The empty leg is reported as a finding about reporting practice: the benchmark with the richest public run-level record publishes no gap in a form whose detectability can even be assessed.

5 Discussion

What changes for a benchmark maintainer. Three practices, each a direct consequence of a table above. (1) Publish the components, or the run-level records from which they are computed: a score without its σ^2_{task} and σ^2_{run} is a measurement without an instrument reading. (2) State the MDE next to the gaps a leaderboard advertises, at the design actually used; the τ-bench retail gap shows the cost of not doing so is not hypothetical. (3) When a target gap is set — "we want to resolve three points" — compute which resource binds before spending on the other: on two of our

three benchmarks the binding resource is tasks, and repetition budgets spent past $R \approx 4$–8 buy almost nothing.

What changes for a reader. A gap below its design's MDE is not evidence of equality; it is an absence of evidence, and the two readings license opposite decisions. The audit's arm-sensitive rows add a second caution: for gaps within a factor of about two of the MDE, the classification itself can depend on the estimator, which is an argument for reporting both estimators' design tables — or for treating near-MDE gaps as unresolved by construction.

Relation to the nearest work. Zhou et al. [21] reach a kindred conclusion — repetition buys little — on one static code benchmark, via a two-component generalizability decomposition and a repetition ceiling. Our design tables say why, in a form that survives the change of setting: repetition buys little *when the task term binds*, and whether it binds is a measurable property of the pair (B4's temperature pair is the counterexample where repetition is exactly the right spend). We could not verify Zhou et al.'s numbers: their advertised outcome tensor is absent from their repository at every ref, and we cite their results as their claims, with the data-availability finding on record. That finding is, in miniature, this paper's ingest rule and its B2/B6 exclusions writ large: in this field, at this moment, *advertised openness is not evidence of openness*, and any metrology that skips artefact-level verification inherits the error. Messing [22] shows interaction terms are real in LLM pipeline uncertainty; we do not model facet interactions, and record that as a limitation rather than absorbing it silently. Miller [8] remains the reference for the task component alone; the present results are what that framework cannot see, because its designs have no repeated runs to see it with.

6 Limitations

1. **Three benchmarks, at the floor.** The passing set is the pre-registered minimum. The benchmarks differ in kind (a task suite, a tool-agent benchmark, a code benchmark's repeated-run corpus), which supports the conjunction's generality argument, but three is three: claims are stated per benchmark and the cross-benchmark claim is the narrow one — the falsifier's set-level reading — not a universal law of agent benchmarks.

2. **One benchmark has two agents.** B3's pooled estimate is intrinsically fragile (a cross-fitted pair ratio), and its indeterminacy is partly a consequence. We report it because excluding a passing benchmark after seeing its outcomes would violate the freeze; the same discipline that admits it also caps what it can certify.

3. **The grader component is unmeasured.** E3 is zero by construction on programmatic grading. On judged benchmarks the grader component could be large (judge biases are documented [19]); identifying it requires re-grading frozen trajectories with judge models — new inference, separately priced, explicitly out of this study's registered scope.

4. **No facet interactions.** Following Messing [22], interactions between facets (e.g. prompt $\times$ judge) may contribute variance our additive split attributes to the named components.

5. **Arm B's MDE is unpaired.** The frozen model arm has no per-pair estimand, so its design tables cannot exploit shared tasks and run uniformly larger; audit rows near the boundary are therefore arm-sensitive in a way a paired model arm would partly resolve. We report the asymmetry rather than repair it post hoc.

6. **Disagreement criterion (c) was not evaluable** (frozen artefact carries pooled ρ only), and B1's prior-sensitivity shift confounds prior and agent-set effects. Both are stated where the numbers are used.

7. **Protocol deviations, declared.** Three are on the census record (repaired 2026-11-10; full entries in the census report's deviation section). (i) The census executed under a newer software stack (PyMC 5.25.1 and companions) than the frozen lockfile pins (PyMC 5.16.2) after host rebuilds, and the deviation was not logged at the time; formal verdicts are read from Arm A, which is stack-insensitive in substance, but exact bit-reproduction of the Arm B fits from the deposited record is not guaranteed. (ii) Arm A reports BCa intervals where the registration specifies percentile bootstrap (declared in Section 3.3; verdicts identical under both). (iii) The SUPPORTED / INDETERMINATE row labels rest on a reporting taxonomy (the interval must lie entirely inside $(0, 1)$) that is not itself in the frozen registration, which freezes only the set-level falsifier; the taxonomy was applied consistently to all six arm–benchmark rows, and the set-level falsifier reading — no benchmark refutes the thesis — does not depend on it.

8. **Static corpora.** HAL's corpus is paused and METR's versions evolve; the findings are statements about the benchmarks' published designs and records, framed as such. B4's corpus is a third-party repeated-run release over SWE-bench Verified tasks, not the benchmark's own leaderboard runs; its audit rows are that paper's published contrasts, labelled as such.

7 Conclusion

On the public record as it stands, the noise in an agent benchmark score is measurable, is first-order, and is mostly unreported. Run noise runs at two-fifths to seven-tenths of task noise where the verdict is readable; the designs in use cannot resolve the gaps they are used to advertise, and the binding constraint is usually the task set, not the run count; and of the benchmarks that advertise the data needed to check any of this, half fail verification at the artefact. None of these conclusions required new experiments — only the decision to treat a benchmark score as a measurement, and to hold its reporting to a measurement's standard. The instruments are cheap, frozen in advance, and deposited with this paper; the design tables are the deliverable a maintainer can act on Monday morning.

Data and code availability

All corpora are public research artefacts: the METR public run-level records; the τ-bench official release (MIT); the Bjarnason et al. repeated-run release (CC-BY-4.0, Hugging Face `ASSERT-KTH/agentic-evals-art` pinned revision). The census code, the frozen pre-registration with its hash chain, the ingest ledger, all 297 fit records, and the JSON artefacts from which every table in this paper is drawn are deposited in the public replication package at `https://github.com/ranadhir19/agent-benchmark-metrology-cen` (its root `validate.py` recomputes the headline census numbers from the deposited artefacts). The HAL corpora (B2/B6) could not be redistributed or read; their exclusion is documented in the ingest ledger.

Declaration of competing interest

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AI use disclosure

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